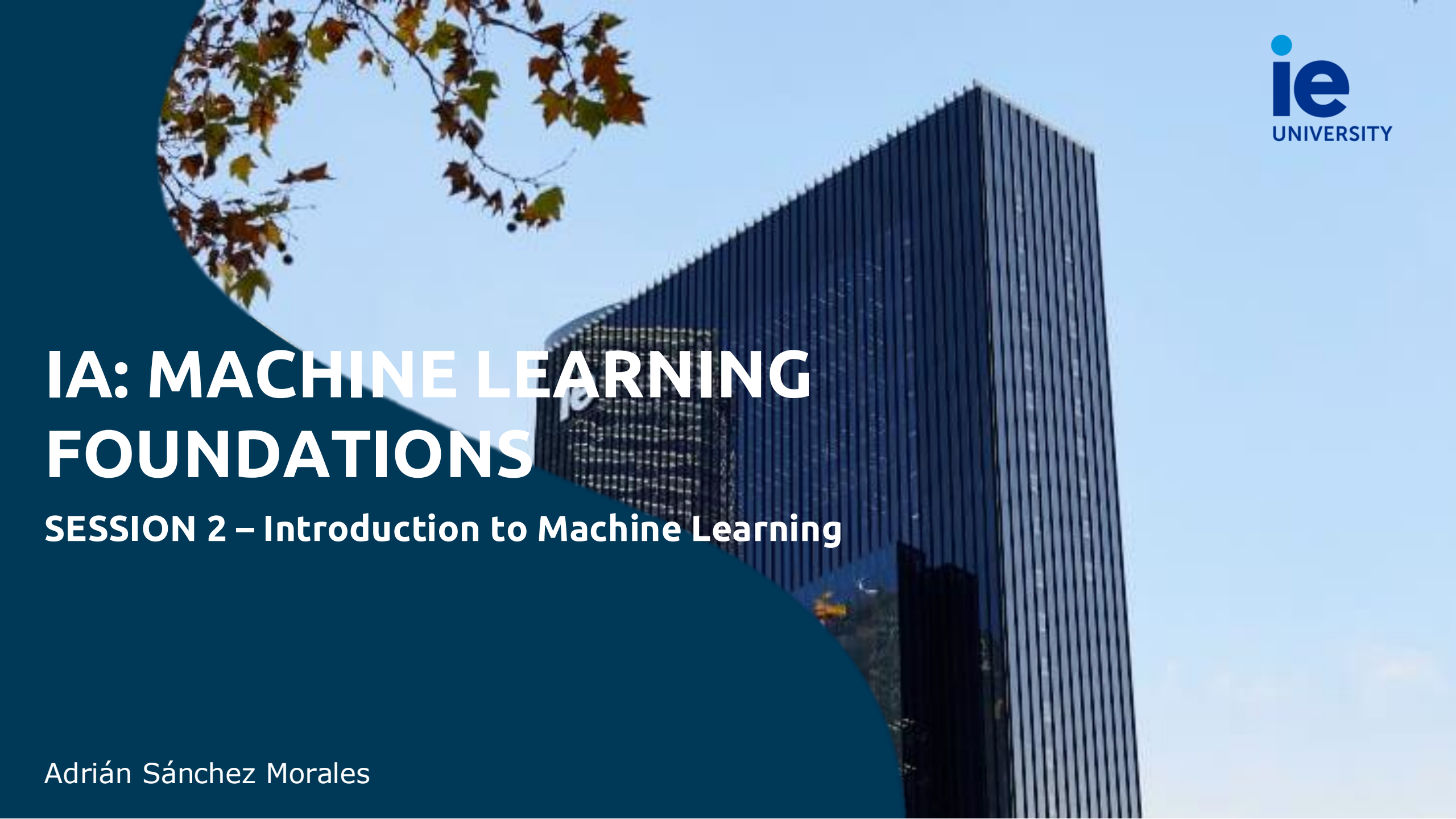


IA: MACHINE LEARNING FOUNDATIONS

SESSION 2 – Introduction to Machine Learning

Adrián Sánchez Morales

AGENDA

1. Introduction
   - 1.1. A bit of history
   - 1.2. What is Statistical Learning

2. The learning problem
   - 2.1. Components of learning
   - 2.2. A simple learning model
   - 2.3. How ML works?

3. Estimation in Machine Learning
   - 3.1. Why estimate f?
   - 3.2. How do we estimate f?
   - 3.3. Trade-off between Accuracy and Interpretability



ie UNIVERSITY Driving Innovation

1. INTRODUCTION

1.1. A BIT OF HISTORY

- At the beginning of the 19th century → Least Squares → Linear Regression (quantitative)
- In 1936 → Linear Discriminant Analysis (quantitative prediction)
- In the 1940s → Logistic Regression
- In the early 1970s → Generalized Linear Model → A set of statistical methods including linear and logistic regression.
- By the end of the 1970s → Many more linear models.
- In the mid 1980s → Classification and regression trees, followed by generalize additive models.
- In the 1980s → Neural Networks
- In the 1990s → Support Vector Machines

1. INTRODUCTION

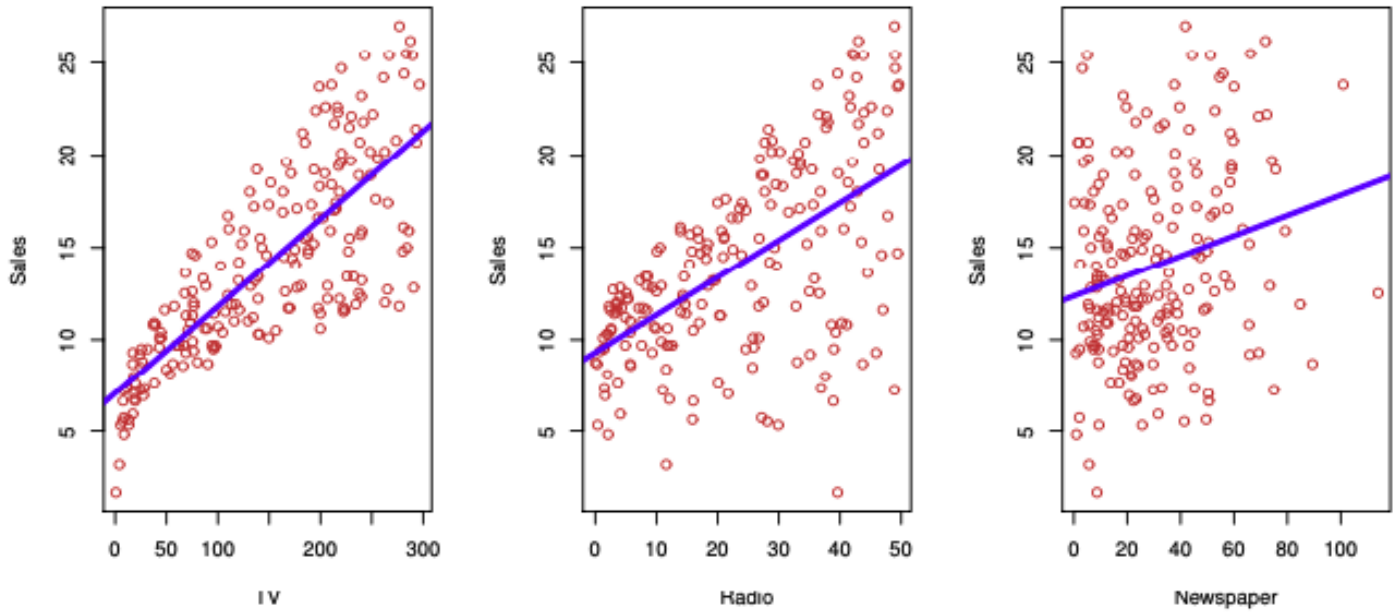
1.1. A BIT OF HISTORY

- Since that time:
  - Statistical learning emerged as a new subfield in statistics
  - Progress of statistical learning increased ⟶ Machine Learning
    - Powerful machines
    - User-friendly software
  - The goal of all of us is to continue the transformation everyday

1. INTRODUCTION

1.2. WHAT IS MACHINE LEARNING?

Advertising data set: Investigate the association between advertising budget and sales of a particular product in 200 different markets.



The client wants to, somehow, control the budget and increase sales.



1. INTRODUCTION

1.2. WHAT IS MACHINE LEARNING?



Develop an accurate model that can be used to predict sales on the basis of the three media budgets.

Input Variables → *Predictors, independent variables, features,* or sometimes just *variables.*

X

Output Variable → *Response* or *dependent variable*

Y

Generally speaking, a relationship is assumed between:

$X = (X_1, X_2, \ldots, X_p)$ and Y

$$Y = f(X) + \epsilon$$

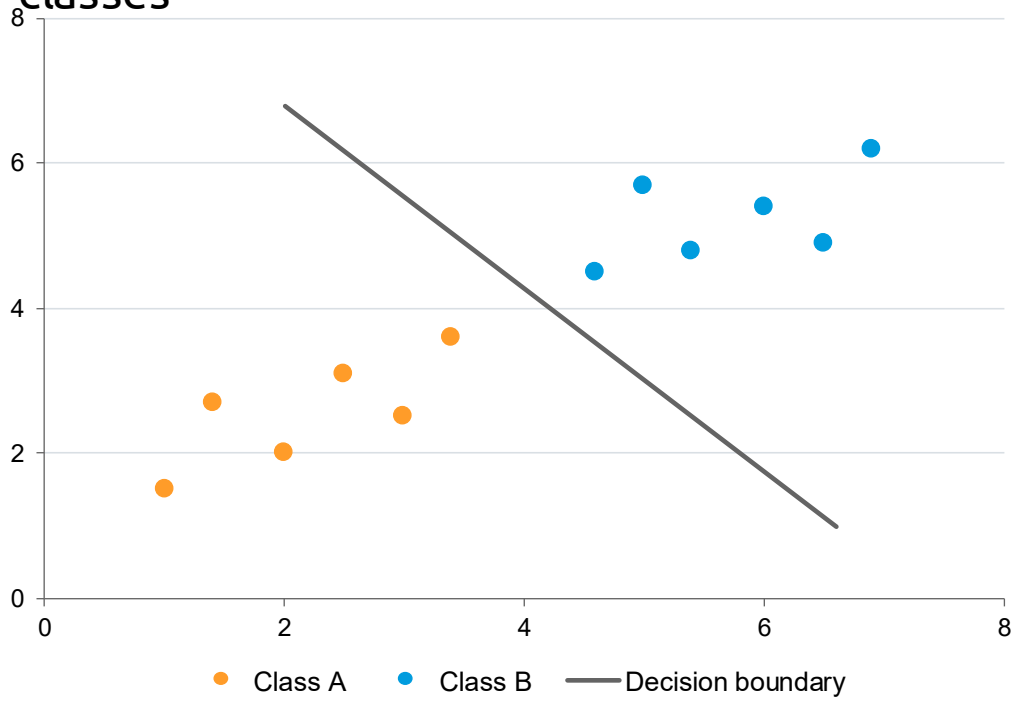
f → Some fixed unknown dependent function

ϵ → Random independent error with zero mean

1. INTRODUCTION

1.2. WHAT IS MACHINE LEARNING?

Classification separates observations into classes



Develop a model that predicts a discrete category from the input features.

Input Variables → *Predictors or features (X)*

Output Variable → *Class label or category (Y)*

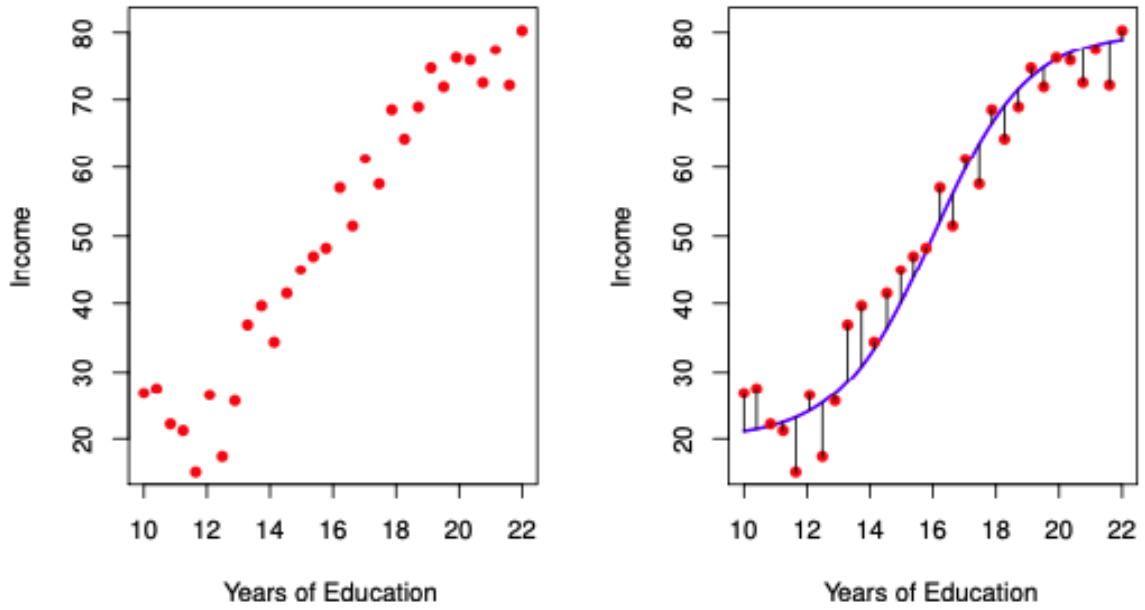
The model learns a decision boundary between classes.

Examples: spam detection · customer churn · medical diagnosis

1. INTRODUCTION

1.2. WHAT IS MACHINE LEARNING?

Income data set



$$Y = f(X) + \epsilon$$

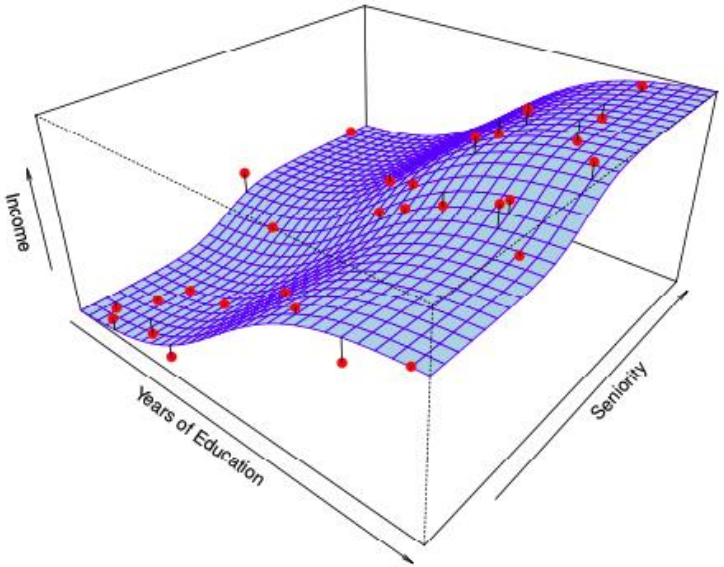
- Nonlinear relationship
- Vertical lines represent the error term with approximately mean zero.

1. INTRODUCTION

1.2. WHAT IS STATISTICAL LEARNING?

Income data set

$$Y = f(X) + \epsilon$$



In essence, statistical learning refers to a set of approaches for estimating f

Machine Learning focuses on developing algorithms and models that allow machines to learn from data to estimate f

AGENDA

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2. THE LEARNING PROBLEM

PROBLEM SETUP

- What does “learn” mean?
- How do we learn that this is a tree?



A perennial plant with an elongated stem, or trunk, supporting leaves or branches



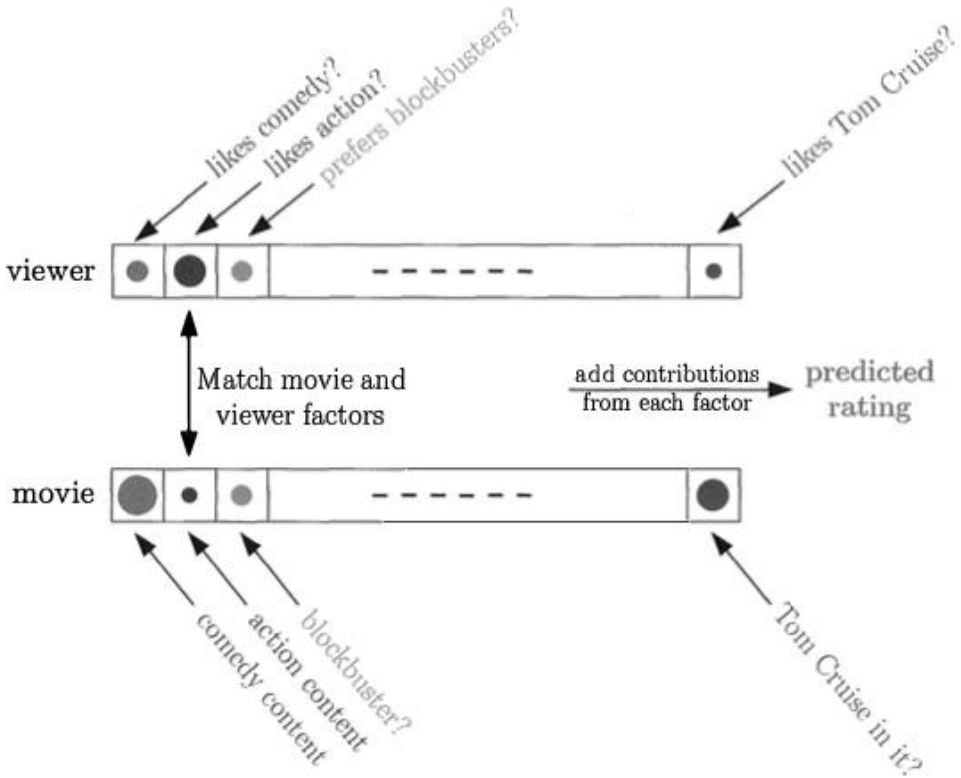
~~Analytic Solution~~ → DATA → Empirical Solution

2. THE LEARNING PROBLEM

PROBLEM SETUP

How users rate movies?

HISTORICAL DATA → Empirical Solution



Approach

- Describe each movie with a number of factors
- Describe each viewer with features

- How a viewer will rate a movie will be estimated depending on the match/mismatch of these factors.

2. THE LEARNING PROBLEM

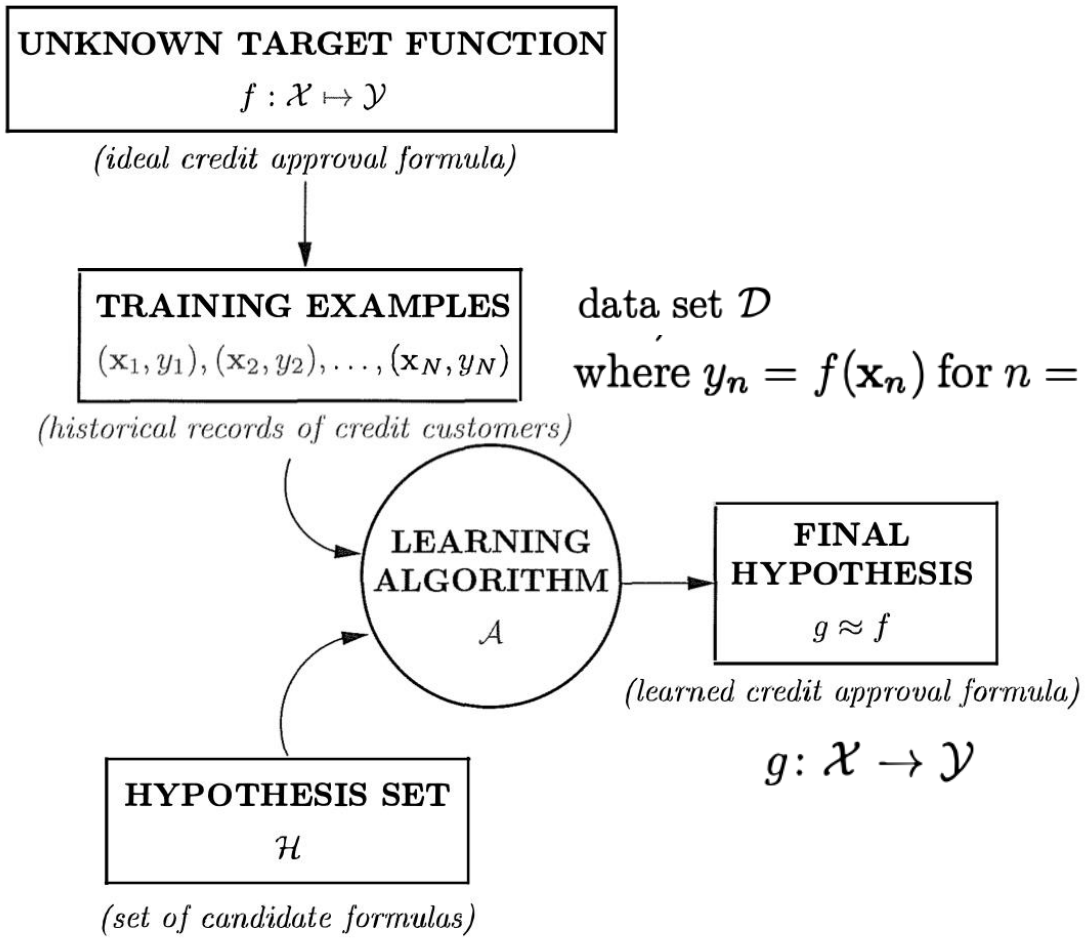
2.1. COMPONENTS OF LEARNING

Imagine a bank receives thousands of credit card applications every day and it wants to automate the process of evaluating them.

Each customer record has personal information:

- Annual salary
- Years in residence
- …
- Credit approved/denied

$$Y = f(X) + \epsilon$$



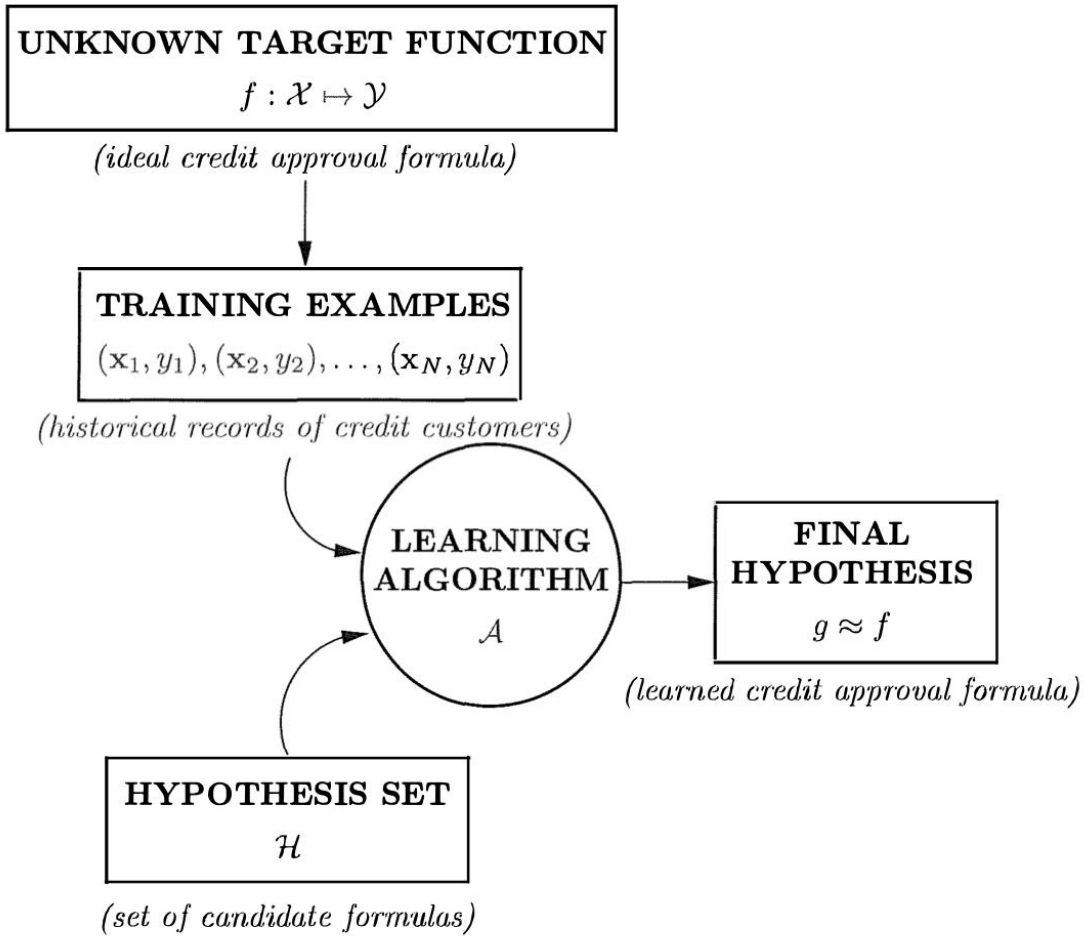
2. THE LEARNING PROBLEM

2.1. COMPONENTS OF LEARNING

EXERCISE

Express each of the following tasks in the framework of learning from data by specifying the input space, output space, target function and the specifics of the data set that we will learn from:

a) Medical diagnosis: A patient walks in with a medical history and some symptoms, and you want to identify the problem.
b) Handwritten digit recognition.
c) Determining if an email is spam or not.
d) Predicting how an electric load varies with prices, temperature and day of the week.



2. THE LEARNING PROBLEM

2.1. COMPONENTS OF LEARNING

SOLUTION

(a)

- X: patient's medical history, symptoms, personal health information etc.
- Y: all possible diseases
- $f: X \rightarrow Y$: ideal formula to identify a patient's problem
- $Dataset$: All available patients' information and their corresponding correct problem diagnostic.

2. THE LEARNING PROBLEM

2.1. COMPONENTS OF LEARNING

SOLUTION

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- X: patient's medical history, symptoms, personal health information etc.
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(b)

- X: handwritten digits (digitalized)
- Y: 0-9 digits
- $f: X \rightarrow Y$: ideal formula match a handwritten digit to a correct digit
- $Dataset$: handwritten digits and their corresponding correct matches

2. THE LEARNING PROBLEM

2.1. COMPONENTS OF LEARNING

SOLUTION

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- X: patient's medical history, symptoms, personal health information etc.
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(c)

- X: every information of an email, e.g. words
- Y: yes/no
- $f: X \rightarrow Y$: ideal formula to identify whether an email is spam or not
- $Dataset$: Spam and non-spam emails that have been identified by human

2. THE LEARNING PROBLEM

2.1. COMPONENTS OF LEARNING

SOLUTION

(a)

- X: patient's medical history, symptoms, personal health information etc.
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- Y: yes/no
- $f: X \rightarrow Y$: ideal formula to identify whether an email is spam or not
- $Dataset$: Spam and non-spam emails that have been identified by human

(d)

- X: price of electric, temperature, day of the week
- Y: electric load
- $f: X \rightarrow Y$: ideal function that gives exact electric load for a given price, temperature and day of the week.
- $Dataset$: historical electric load along with corresponding price, temperature and day of the week information.

2. THE LEARNING PROBLEM

2.2. A SIMPLE LEARNING MODEL

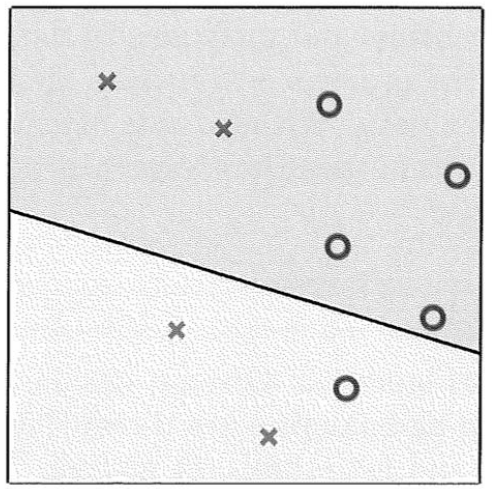
$$\mathcal{X} = \mathbb{R}^d$$

$$\mathcal{Y} = \{+1, -1\}$$

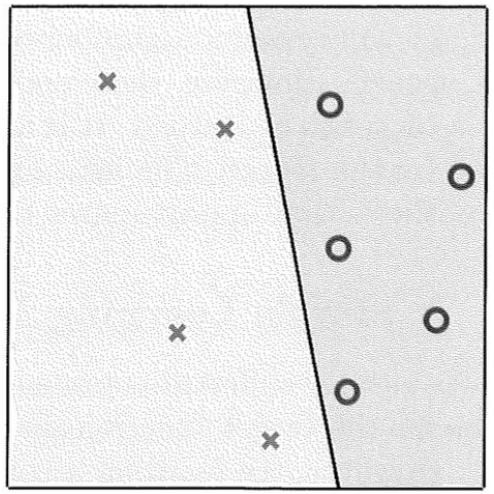
Approve credit if $\sum_{i=1}^{d} w_i x_i > \text{threshold}$

Deny credit if $\sum_{i=1}^{d} w_i x_i < \text{threshold}$

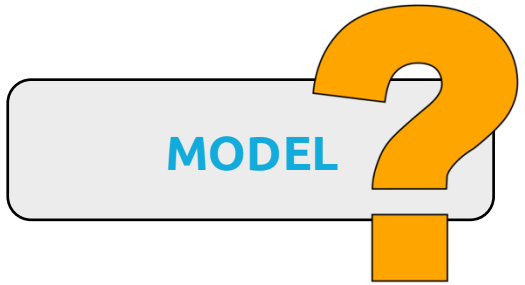
$$h(\mathbf{x}) = \text{sign}\left(\left(\sum_{i=1}^{d} w_i x_i\right) + b\right)$$



(a) Misclassified data



(b) Perfectly classified data



2. THE LEARNING PROBLEM

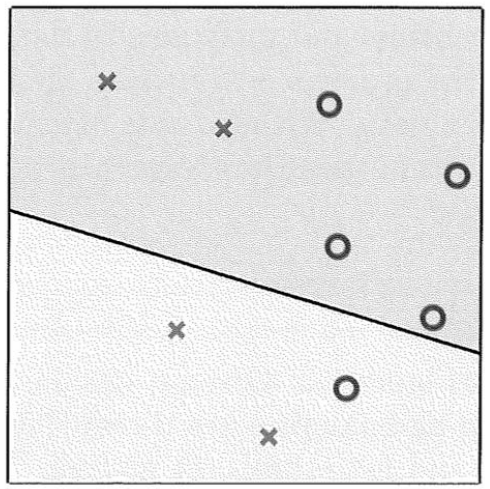
2.2. A SIMPLE LEARNING MODEL

$$\mathcal{X} = \mathbb{R}^d \qquad \mathcal{Y} = \{+1, -1\}$$

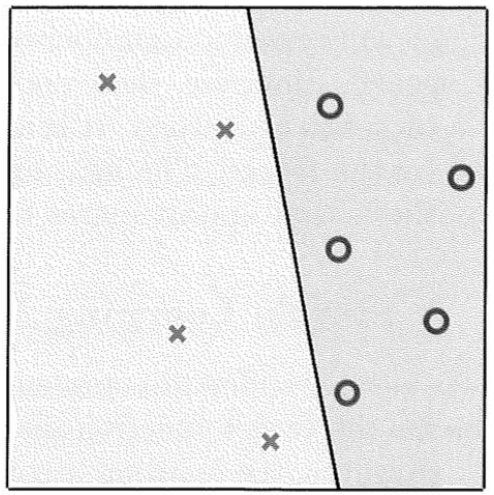
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$$h(\mathbf{x}) = \text{sign}\left(\left(\sum_{i=1}^{d} w_i x_i\right) + b\right)$$



(a) Misclassified data



(b) Perfectly classified data

PERCEPTRON

2. THE LEARNING PROBLEM

2.2. A SIMPLE LEARNING MODEL

EXERCISE

Suppose that we use a perceptron to detect spam messages. Let's say that each email message is represented by the frequency of occurrence of keywords, and the output is +1 if the message is considered spam:

a) Think of some keywords that will end up with a large positive weights?
b) How about keywords that will get a negative weight?
c) What parameter in the perceptron directly affects how many borderline messages end up being classified as spam?

2. THE LEARNING PROBLEM

2.2. A SIMPLE LEARNING MODEL

EXERCISE

Suppose that we use a perceptron to detect spam messages. Let's say that each email message is represented by the frequency of occurrence of keywords, and the output is +1 if the message is considered spam:

a) Think of some keywords that will end up with a large positive weights?
b) How about keywords that will get a negative weight?
c) What parameter in the perceptron directly affects how many borderline messages end up being classified as spam?

SOLUTION

(a) Keywords with a large positive weight: free, cheap, earn, !

(b) Keywords with a negative weight: person name, hi, the

(c) The parameter b in perceptron directly affects how many borderline messages end up being classified as spam. This is because b is the threshold used to classify the emails into spam and non-spam categories.

2. THE LEARNING PROBLEM

2.3. HOW ML WORKS?

Phases of a Machine Learning model

The process of building an analytical model is generally based on four well-defined phases:



1

Data Processing

Collect and format a large amount of data



2

Training and Validation

Teach the model to obtain the expected result and validate that it works.



3

Testing & Deployment

Present data not previously seen by the model and put into a production environment.



4

Maintenance

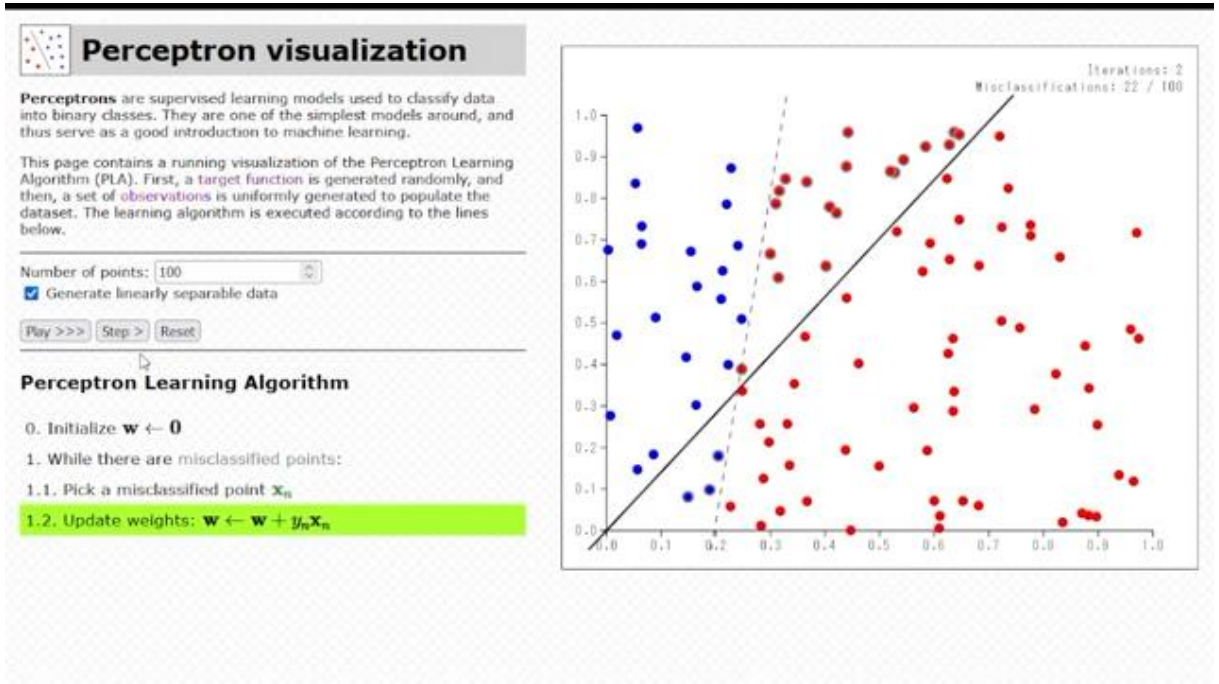
Monitor the model to ensure that its behavior is not degraded.

2. THE LEARNING PROBLEM

2.3. HOW ML WORKS?

All AIs are created from **mathematical models** that must be **trained** in order to fulfill their function. But what does the training of these models consist of?

- The process by which a Machine **Learning algorithm learns** to perform specific tasks from data.
- It is what allows a model to make predictions, classifications or decisions based on new and previously unseen data.
- Avoid GIGO (Garbage In – Garbage Out):
  - The quality of the data on which a model is trained determines its effectiveness
  - If they are wrong, incomplete, or biased, the model will replicate these problems by making poor, incorrect, or biased predictions



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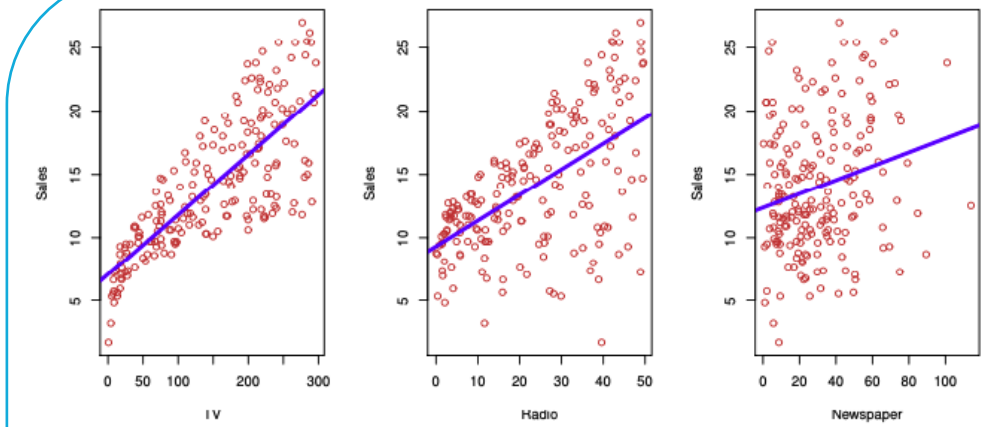
3. ESTIMATION IN MACHINE LEARNING

3.1. WHY ESTIMATE f?

Inference (Data Analysis) – Explains relationships

Sometimes, we may be interested in understanding the association between Y and $(X_1\ X_2, X_3\ \ldots X_p)$. We may be interested in answering questions like:

- Which predictors are associated with the response?
- What is the relationship between the response and each predictor?
- Can the relationship between Y and each predictor be adequately summarized using a linear equation, or is the relationship more complicated?



- Which media are associated with sales?
- Which media investment is most strongly associated with sales?
- How large of an increase in sales is associated with a given increase in TV advertising?



¿ Association = causation ?

3. ESTIMATION IN MACHINE LEARNING

3.1. WHY ESTIMATE f?

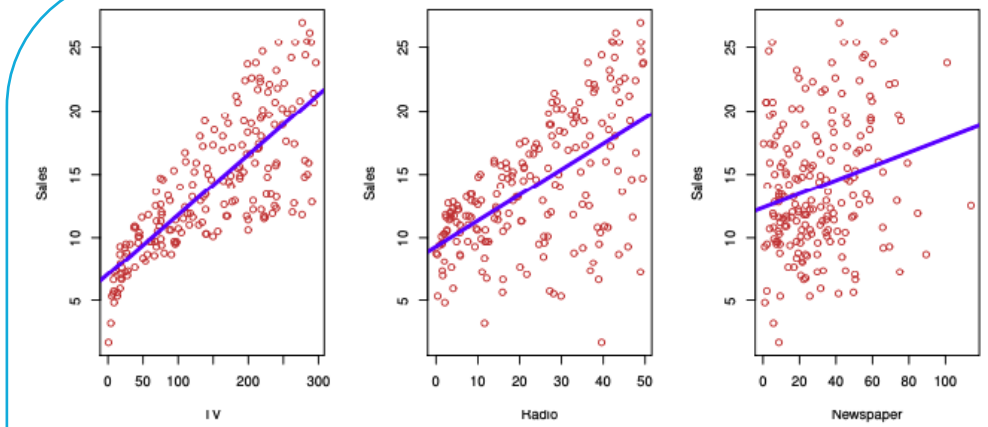
Inference (Data Analysis) – Explains relationships

Association ≠ causation

Observational data can reveal relationships, but not necessarily causal effects. Establishing causality requires stronger assumptions or experimental designs, such as randomized controlled trials.

For example, higher TV advertising spending may coincide with higher sales, but just this does not prove that advertising caused the increase. The relationship could be explained by:

- **Confounding variables**: seasonality, promotions, prices, or prior demand.
- **Reverse causality**: the company increases advertising because it anticipates higher sales.
- **Selection bias**: more is invested in markets that already have better prospects.



- Which media are associated with sales?
- Which media investment is most strongly associated with sales?
- How large of an increase in sales is associated with a given increase in TV advertising?

3. ESTIMATION IN MACHINE LEARNING

3.1. WHY ESTIMATE f?

Prediction (Forecasting & Classification) – Optimizes accuracy on unseen data

Objective: $Y = f(X) + \epsilon$ ⟶ Our estimate: $\hat{Y} = \hat{f}(X)$

$$\hat{Y} = reducible\ error + irreducible\ error$$

- *Reducible* because we can potentially improve the accuracy of $\hat{y}$ by improving the algorithm.
- *Irreducible* part because Y is also a function of ϵ and it cannot be reduced.

Why is the irreducible error larger than zero?

- Unmeasured variables
- Unmeasured variation

EXAMPLE

Estimate severity of a patient's reaction to a drug having blood information.

The risk may vary for a given patient on a given day.

- Manufacturing variation in the drug
- Patient's general feeling
- Well-being that day

3. ESTIMATION IN MACHINE LEARNING

3.2. HOW DO WE ESTIMATE f?

Parametric Methods

Step I

We make an assumption about the functional form, or shape, of f:

$$f(X) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p.$$

Thus, the problem is simplified by just having to estimate the coefficients.

Step II

After a model has been selected, we need a procedure that uses the training data to fit or train the model.

$$Y \approx \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p.$$

(Ordinary) Least Squares

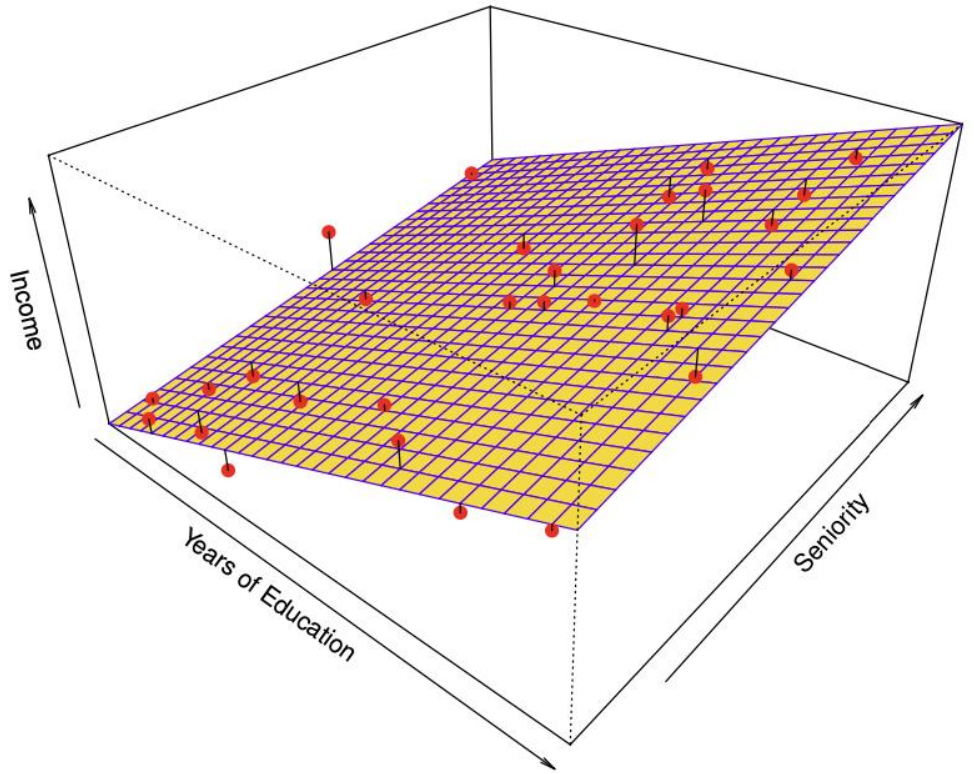
3. ESTIMATION IN MACHINE LEARNING

3.2. HOW DO WE ESTIMATE f?

Parametric Methods

- Model-based approach
- The problem of estimating f is simplified since it is easier to estimate some parameters than the it is to fit an entirely arbitrary function f.
- **DISADVANTAGE**: If the chosen model is too far from the true f, then our estimate will be poor.
- We can try to solve this by using more *flexible* models, but:
  - More complex
  - Overfitting

$$\texttt{income} \approx \beta_0 + \beta_1 \times \texttt{education} + \beta_2 \times \texttt{seniority}$$



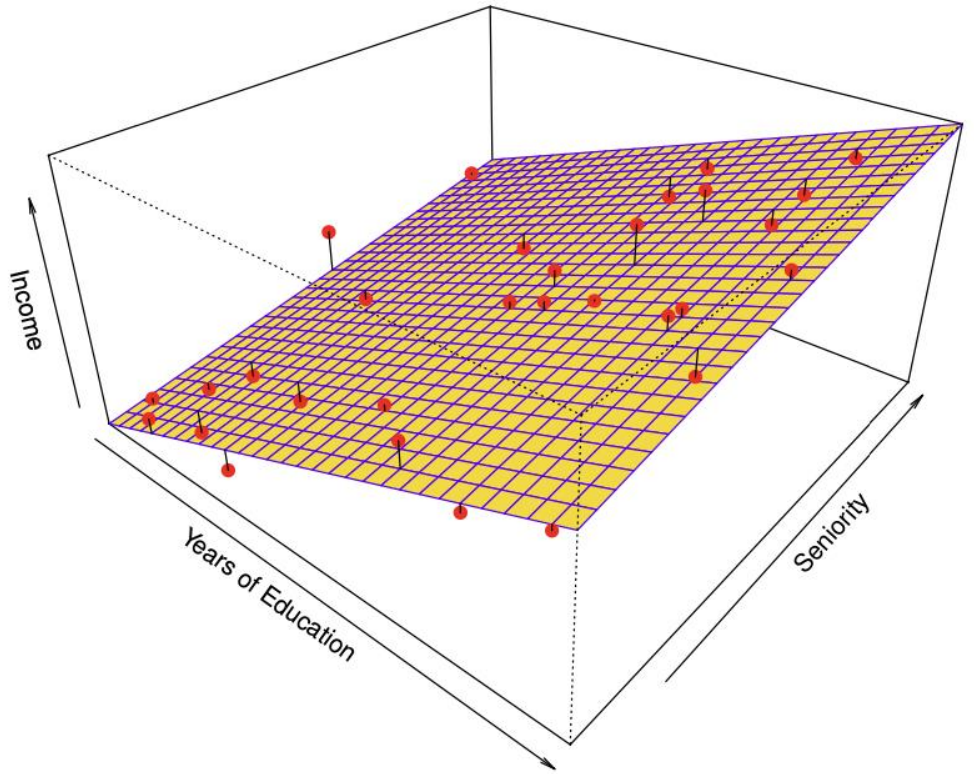
3. ESTIMATION IN MACHINE LEARNING

3.2. HOW DO WE ESTIMATE f?

Parametric Methods

$$\texttt{income} \approx \beta_0 + \beta_1 \times \texttt{education} + \beta_2 \times \texttt{seniority}$$

- Linear Regression
- Logistic Regression
- Linear Discriminant Analysis
- Perceptron (NN with fixed architecture)
- Naive Bayes

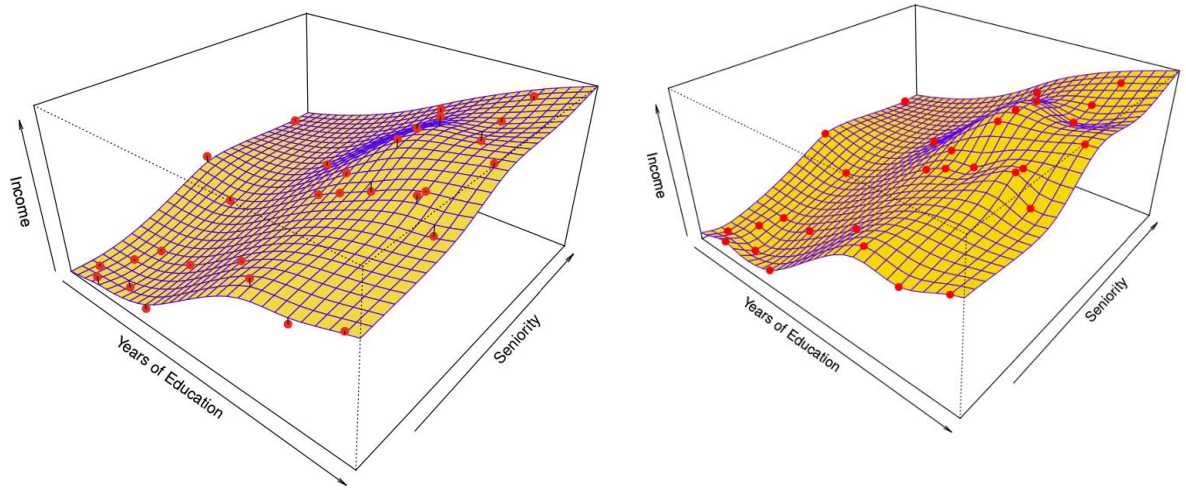


3. ESTIMATION IN MACHINE LEARNING

3.2. HOW DO WE ESTIMATE f?

Non-Parametric Methods

- Non-parametric methods do not make explicit assumptions about the functional form of f.
- They seek an estimate of f that gets as close to the data points as posible.
- **DISADVANTAGE**: since they do not reduce the problem of estimating f to a small number of parameters, a very large number of observations is required in order to obtain an accurate estimate.

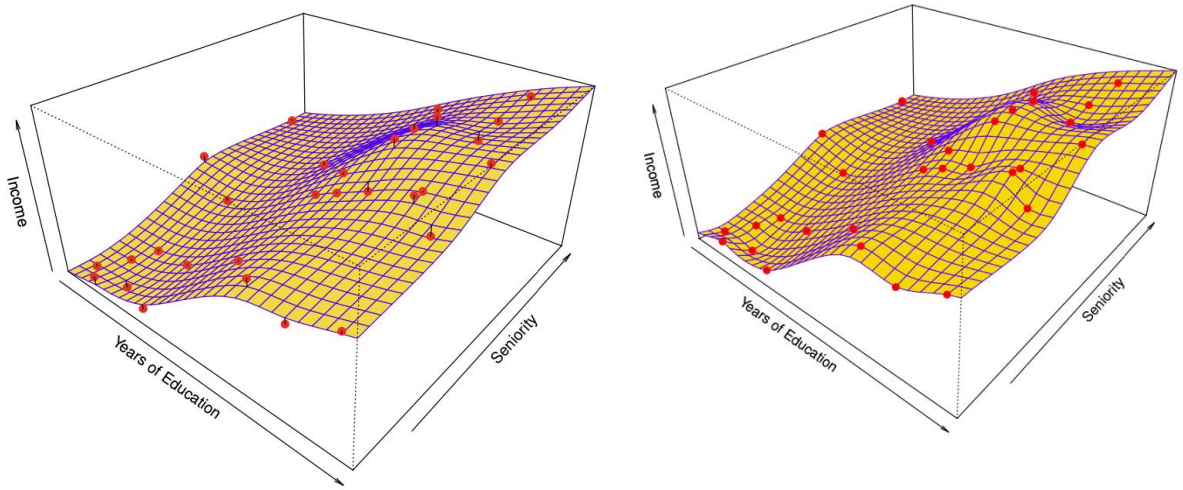


3. ESTIMATION IN MACHINE LEARNING

3.2. HOW DO WE ESTIMATE f?

Non-Parametric Methods

- k-Nearest Neighbors
- Decision Trees like CART and C4.5
- Support Vector Machines
- MLP



3. ESTIMATION IN STATISTICAL LEARNING

3.2. HOW DO WE ESTIMATE f?

Aspect	Parametric Methods	Non-Parametric Methods
Assumptions	Assume a specific data distribution (e.g., normal)	No strict assumptions about data distribution
Number of Parameters	Fixed	Grows with dataset size
Flexibility	Less flexible	Highly flexible
Computational Cost	Lower	Higher for large datasets
Data Requirements	Interval or ratio scale	Works with ordinal or non-normal data
Examples (ML)	Linear Regression, Logistic Regression, Naïve Bayes	K-Nearest Neighbors, Decision Trees, SVM with kernels

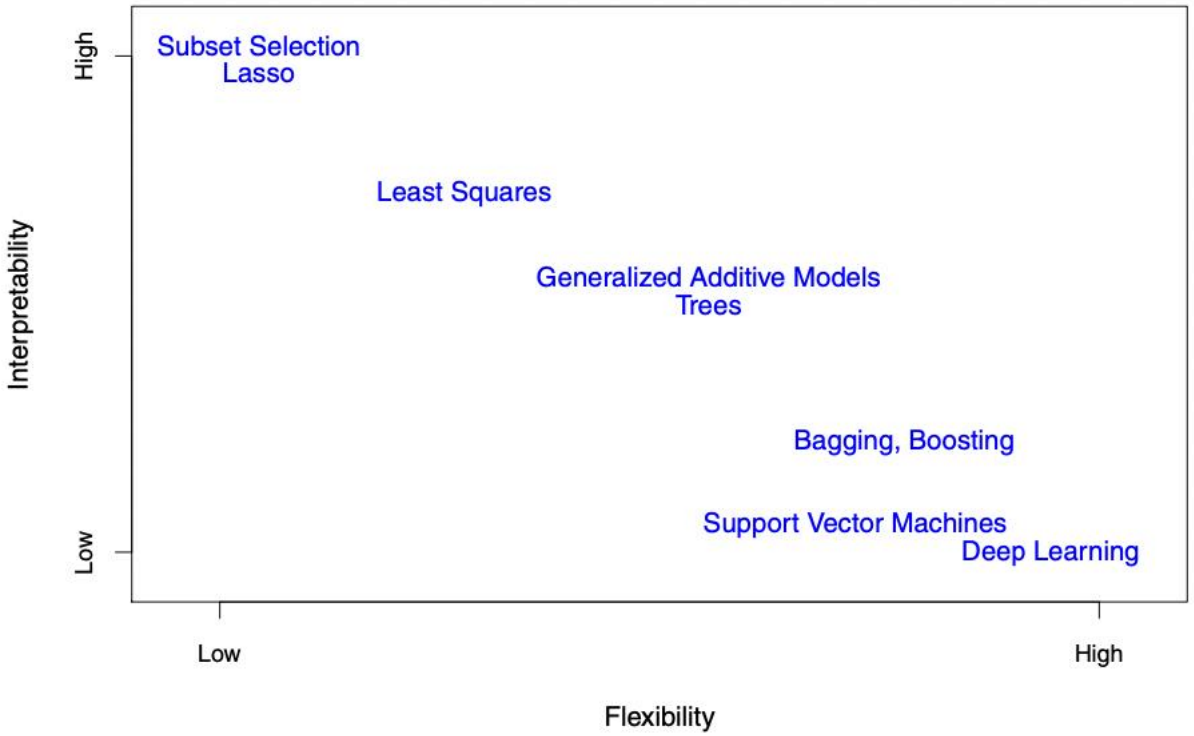
3. ESTIMATION IN MACHINE LEARNING

3.3. ACCURACY vs INTERPRETABILITY

- Flexible model: can generate a much wider range of possible shapes to estimate f.

- *Why would we ever choose to use a more restrictive method?*

Interpretability



LINEAR MODEL

$$Y \approx \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p.$$


THANK YOU

PYTHON ENVIRONMENT

FROM SCRATCH (WINDOWS)

1. Download Python Installer – python.org
2. Run the executable installer
3. Add Python to environmental variables
4. Install pip
5. Create environment manually
6. Install every library manually with pip

*** NOTE:** Be careful with the operating system of the laptop.

PYTHON ENVIRONMENT

ANACONDA



Data science technology for
a better world.

Anaconda offers the easiest way to perform Python/R data science and machine learning on a single machine. Start working with thousands of open-source packages and libraries today.

https://www.anaconda.com